UK Plastic Packaging Tax – ESG Compliance Engine
End-to-End ESG Data Engineering & Analytics Proof of Concept

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1. Introduction

1.1 Project Purpose:

This project presents a **Proof of Concept (PoC)** that demonstrates how modern data engineering techniques can be used to integrate **transactional retail data** with **external sustainability metadata** in order to quantify financial exposure to the **UK Plastic Packaging Tax**.

In most enterprise environments, transactional systems (e.g., POS, ERP) are designed to capture:

- Revenue
- Product identifiers
- Sales volumes

However, these systems typically lack **environmental attributes**, such as:

- Packaging material composition
- Recyclability
- Sustainability indicators

This creates a structural disconnect:

Transactional systems provide financial data, while sustainability data exists externally.

The core objective of this project is therefore:

To bridge the gap between financial and ESG data domains and translate sustainability attributes into measurable financial impact.

1.2 Problem Statement:

The project addresses the following technical and business challenge:

"How can we enrich high-volume transactional sales data with external ESG metadata to compute regulatory tax exposure in a scalable and auditable manner?"

This involves:

- Data ingestion and integration
- ESG enrichment logic
- Financial modelling
- Analytical reporting

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2. Business Context

2.1 UK Plastic Packaging Tax Overview:

The UK Plastic Packaging Tax is a regulatory mechanism designed to incentivise the use of recycled materials and reduce environmental impact.

Key characteristics:

- Tax rate: **£223.69 per tonne of plastic**
- Applies to plastic packaging components
- Triggered when recycled content thresholds are not met

Business Implications:

Organisations must:

- Identify plastic usage in packaging
- Calculate tax liability
- Maintain audit-ready reporting

Failure to comply can result in:

- Financial penalties
- Regulatory scrutiny
- Reputational damage

2.2 Business Problem Definition:

Retail organisations face a fundamental data limitation:

Available Data	Missing Data
Sales transactions	Packaging material
Revenue	Plastic weight
Product IDs	ESG attributes

As a result:

- Tax liability cannot be computed directly
- ESG reporting is incomplete
- Risk exposure is unknown

This project introduces a **data engineering solution** to overcome this limitation using:

- External ESG datasets
- Heuristic modelling
- Data warehouse transformations

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3. Data Sources

3.1 Retail Star Schema Dataset:

The primary dataset represents a classical **star schema**:

Fact Table:

- FACT_SALES_RAW
- ~1,000,000 rows
- Contains:
- sales_id
- product_sk
- sales_date
- total_amount

Dimension Tables:

- DIM_PRODUCTS_RAW
- DIM_DATES_RAW

Characteristics:

- Structured and normalized
- High volume
- No ESG attributes

3.2 Amazon Eco-Friendly Dataset:

This dataset provides **external sustainability metadata**, including:

- Product title
- Category
- Material description
- Sustainability keywords

Purpose in this project:

Used as a:

Reference dataset for ESG enrichment

It enables:

- Identification of eco-friendly materials
- Mapping of sustainability signals
- Approximation of packaging characteristics

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4. Architecture Overview

The system follows a **Medallion Architecture** implemented in Snowflake:

```
ESG_PL_DB
│
├── ESG_RAW    → Raw ingestion (Bronze)
├── ESG_SILVER → Transformation & enrichment
└── ESG_GOLD   → Analytical layer
```

Rationale:

- Separation of concerns
- Traceability
- Scalability
- Reusability of transformations

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5. Bronze Layer (ESG_RAW)

5.1 Objective:

The Bronze layer is designed to:

- Store data **exactly as received**
- Maintain an audit trail
- Avoid premature transformations

5.2 Implementation:

Database Setup:

```
CREATE DATABASE ESG_PL_DB;
CREATE SCHEMA ESG_PL_DB.ESG_RAW;
CREATE SCHEMA ESG_PL_DB.ESG_SILVER;
CREATE SCHEMA ESG_PL_DB.ESG_GOLD;
```

File Format Configuration:

```
CREATE OR REPLACE FILE FORMAT CSV_FMT
TYPE = CSV
FIELD_OPTIONALLY_ENCLOSED_BY = '"'
SKIP_HEADER = 1;
CREATE OR REPLACE FILE FORMAT PARQUET_FMT
TYPE = PARQUET;
```

Stage Creation:

```
CREATE OR REPLACE STAGE ESG_RAW_STAGE
FILE_FORMAT = CSV_FMT;
```

Data Loading Example:

```
COPY INTO FACT_SALES_RAW
FROM @ESG_RAW_STAGE/fact_sales_normalized.csv
FILE_FORMAT = (FORMAT_NAME = CSV_FMT)
ON_ERROR = 'CONTINUE';
```

5.3 Validation:

Validation checks included:

```
SELECT COUNT(*) FROM FACT_SALES_RAW;
SELECT COUNT(*) FROM DIM_PRODUCTS_RAW;
SELECT COUNT(*) FROM DIM_DATES_RAW;
```

Expected:

- ~1,000,000 sales records
- ~210 products
- ~366 dates

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6. Silver Layer (ESG_SILVER)

6.1 Objective:

The Silver layer transforms raw data into **ESG-enriched transactional records**.

6.2 Data Enrichment Logic:

6.2.1 Product Join:

```
FROM FACT_SALES_RAW f
JOIN DIM_PRODUCTS_RAW p
  ON f.product_sk = p.product_sk
```

This attaches:

- Product name
- Category
- Brand

6.2.2 Plastic Weight Estimation:

Since real packaging data is unavailable, a **heuristic model** was used:

```
CASE
  WHEN LOWER(p.category) LIKE '%home%' THEN 0.25
  WHEN LOWER(p.category) LIKE '%grocery%' THEN 0.15
  WHEN LOWER(p.category) LIKE '%electronics%' THEN 0.10
  WHEN LOWER(p.category) LIKE '%clothing%' THEN 0.05
  ELSE 0.08
END AS plastic_weight_kg
```

Justification:

- Represents typical packaging intensity
- Enables modelling without supplier data
- Common approach in early-stage ESG analytics

6.2.3 ESG Classification:

```
CASE
  WHEN LOWER(p.category) LIKE '%grocery%' THEN 'MATCHED'
  ELSE 'UNCLASSIFIED'
END AS esg_match_status
```

Purpose:

- Distinguish products with inferred ESG alignment
- Identify data gaps explicitly

6.2.4 Governance Flag (Critical Design Decision):

```
CASE
  WHEN esg_match_status = 'UNCLASSIFIED' THEN 1
  ELSE 0
END AS is_unclassified
```

Why this is important

- Avoids string filtering issues in BI
- Ensures consistent aggregation
- Improves model reliability

6.3 Output Table:

ESG_SILVER.SALES_ENRICHED

Contains:

- Sales data
- Plastic weight
- ESG classification
- Governance flags

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7. Gold Layer (ESG_GOLD)

7.1 Objective:

Provide **analytics-ready datasets** for business intelligence tools.

7.2 Tax Calculation:

```
(projected_tax_liability_gbp) = (plastic_weight_kg / 1000) * 223.69
```

Explanation:

- Convert kg → tonnes
- Multiply by tax rate

7.3 FACT_COMPLIANCE View:

```
CREATE OR REPLACE VIEW FACT_COMPLIANCE AS
SELECT
   sales_id,
   product_id,
   category,
   total_amount,
   plastic_weight_kg,
   is_unclassified,
   (plastic_weight_kg / 1000) * 223.69 AS projected_tax_liability_gbp,
   DATE_FROM_PARTS(year, month, 1) AS year_month_date
FROM ESG_SILVER.SALES_ENRICHED;
```

7.4 CATEGORY_RISK:

```
SELECT
  category,
  COUNT(*) AS sales_count,
  SUM(projected_tax_liability_gbp) AS total_tax
FROM FACT_COMPLIANCE
GROUP BY category;
```

7.5 MONTHLY_ESG_TREND:

Aggregates:

- Monthly plastic usage
- Monthly tax
- Revenue

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8. Power BI Semantic Layer

8.1 Measures:

Revenue:

```
Total Revenue = SUM(FACT_COMPLIANCE[TOTAL_AMOUNT])
```

Tax:

```
Total Plastic Tax =
SUM(FACT_COMPLIANCE[PROJECTED_TAX_LIABILITY_GBP])
```

Tax %:

```
Plastic Tax % of Revenue = DIVIDE([Total Plastic Tax], [Total Revenue])
```

Density:

```
Plastic Density = DIVIDE( SUM(FACT_COMPLIANCE[PLASTIC_WEIGHT_KG]),
COUNT(FACT_COMPLIANCE[SALES_ID])
)
```

Unclassified:

```
Unclassified Products = SUM(FACT_COMPLIANCE[IS_UNCLASSIFIED])
```

8.2 Dashboard Visualisation (Power BI)

The final stage of this project involved building an interactive Power BI dashboard to present ESG compliance insights derived from the Gold data layer.

The dashboard translates complex data transformations into intuitive visual analytics, enabling stakeholders to monitor financial exposure, ESG classification gaps, and operational efficiency in packaging usage.

It integrates multiple analytical perspectives:

- Financial performance
- Regulatory tax exposure
- Category-level risk distribution
- ESG data completeness

Figure 1: ESG Compliance Dashboard

8.3 Dashboard Components Explanation:

Financial KPIs

The dashboard presents three primary KPIs:

- Total Revenue (~£2.75bn)
- Total Plastic Tax (~£24k)
- Plastic Tax % of Revenue (~0.00087%)

These metrics provide an executive-level overview of financial exposure and demonstrate that, while current tax impact is low, it remains directly tied to sales volume and packaging assumptions.

Plastic Density Trend (kg per sale):

This time-series visual tracks the average plastic weight per transaction over time.

- The trend appears relatively stable
- No consistent downward pattern is observed

This indicates that:

- Packaging optimisation initiatives are not yet evident
- Sustainability improvements are not embedded operationally

Plastic Tax by Category:

This visual highlights the distribution of tax liability across product categories.

Key observations:

- Home Appliances and Electronics contribute the highest tax
- Tax exposure is unevenly distributed

This reveals:

- Category concentration risk
- High-impact areas for cost optimisation

Unclassified Products by Category:

This visual represents the volume of products lacking ESG classification.

Key findings:

- A significant number of products remain unclassified (~833k)
- Distribution is spread across multiple categories

This indicates:

- Data governance gaps
- Potential compliance risks
- Lack of supplier-level ESG transparency

The dashboard demonstrates how raw transactional and ESG data can be transformed into actionable financial insights, enabling data-driven sustainability and compliance decision-making

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9. Insights (Detailed Analysis)

9.1 Financial Insights:

The analysis reveals that the organisation generates approximately **£2.75 billion in annual revenue**, while the estimated **plastic packaging tax liability is around £24,000**, resulting in a **tax-to-revenue ratio of approximately 0.00087%**.

At first glance, this suggests that the current financial impact of the plastic tax is **immaterial at the aggregate level**. However, this interpretation must be treated with caution.

Key Interpretation:

The low ratio is primarily driven by:

- **Small per-unit tax cost** (fractions of a penny per transaction)
- **Heuristic plastic weight assumptions**
- The current **tax rate being applied uniformly**

However, from a financial modelling perspective, this represents a **baseline scenario**, not a future-proof estimate.

Strategic Insight:

Although the current exposure is minimal:

- The tax is **volume-dependent**, meaning it scales linearly with sales growth
- Regulatory changes (e.g., increased tax rate or expanded scope) could significantly increase exposure
- Categories with higher packaging intensity will **disproportionately amplify cost increases**

Business Implication:

The organisation currently operates in a “low visible cost, high latent risk” position.

This means:

- The tax is not yet impacting margins materially
- But future regulatory tightening could turn this into a **non-trivial cost centre**

9.2 Risk Insights:

9.2.1 Category Concentration Risk:

The analysis shows that tax exposure is **not evenly distributed across categories**, but instead **concentrated in a small subset of product categories**, particularly:

- Home Appliances
- Electronics
- Furniture

Why this matters:

These categories typically:

- Use **heavier packaging materials**
- Have **complex supply chains**
- Require protective packaging due to product fragility

As a result:

- They generate **higher plastic weight per unit**
- They contribute disproportionately to tax liability

Risk Interpretation:

This creates a **concentration risk**, where:

- A small number of categories drive the majority of tax exposure
- Any regulatory change will impact these categories first and most severely

Business Impact:

- Reduced pricing flexibility (especially in competitive markets)
- Margin compression in high-volume categories
- Increased pressure on procurement and supplier negotiations

9.2.2 ESG Data Gap Risk (Unclassified Products):

The dataset shows approximately **833,000 unclassified transactions**, representing a significant portion of total sales.

Interpretation:

Unclassified products are those where:

- ESG attributes could not be inferred
- Material composition is unknown
- Sustainability signals are missing

Why this is critical:

From a regulatory perspective:

- Missing data is often treated conservatively
- Organisations may be required to assume **worst-case scenarios**

Risk Exposure:

- Overestimation of tax liability during audits
- Reduced confidence in ESG reporting
- Increased compliance risk

Technical Insight:

This highlights a key data engineering challenge:

"Absence of data is itself a risk signal."

Business Impact:

- Compliance teams lack visibility
- Finance teams cannot accurately forecast tax exposure
- ESG reporting may be challenged during audits

9.3 Operational Insights:

9.3.1 Plastic Density Trend:

The analysis of **plastic density (kg per sale)** over time shows a **relatively stable trend**, with no significant downward movement.

Interpretation:

This indicates:

- Packaging practices have **not materially improved**
- Sustainability initiatives are either:
- Not implemented
- Not measurable in current data

Why this matters:

A declining trend would indicate:

- Packaging optimisation
- Material substitution
- ESG maturity

However, a flat trend suggests:

- Operational inertia
- Lack of measurable sustainability KPIs

Operational Risk:

- The organisation is not actively reducing plastic intensity
- Future regulatory tightening will directly increase costs
- No early mitigation strategy is visible

Strategic Insight:

Plastic reduction is not yet embedded as an operational metric.

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10. Limitations (Critical Evaluation)

This section is **very important academically** — it shows you understand the boundaries of your solution.

10.1 Heuristic-Based Modelling:

Plastic weight is derived using **category-based assumptions**, rather than actual measurements.

Limitation:

- Does not reflect real packaging variation
- Ignores:
- Product size differences
- Supplier packaging standards
- Regional variations

Impact:

- Tax estimates are approximate
- Results should be interpreted as **indicative, not exact**

10.2 Absence of Real Packaging Data:

The system does not include:

- Supplier-provided packaging specifications
- Bill of materials (BOM)
- Recycled content percentages

Impact:

- Cannot determine true tax eligibility
- Cannot validate compliance thresholds
- Reduces accuracy of ESG classification

10.3 Static Tax Rate:

The tax rate is fixed at:
£223.69 per tonne
Limitation:

- Does not account for:
- Future rate increases
- Policy changes
- Regional tax differences

Impact:

- No forward-looking scenario modelling
- Limited strategic planning capability

10.4 Simplified ESG Mapping:

ESG classification is based on:

- Category keywords
- Basic logic rules

Limitation:

- Does not capture nuanced sustainability attributes
- Cannot detect:
- Misleading product descriptions
- Mixed-material packaging
- Supplier claims accuracy

Summary of Limitations:

Area	Limitation	Impact
Data	Missing packaging details	Reduced accuracy
Logic	Heuristic assumptions	Approximate results
Model	Static tax rate	No forecasting
ESG	Simplified classification	Limited depth

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11. Future Enhancements

This section shows **how you would productionise this system**.

11.1 Supplier-Level Packaging Integration:

Proposal:

Integrate:

- Supplier-provided packaging data
- Material composition (plastic %, recycled %)

Benefits:

- Accurate tax calculation
- Compliance validation
- Improved ESG reporting

11.2 NLP-Based ESG Classification:

Proposal:

Use Natural Language Processing (NLP) to analyse:

- Product descriptions
- Material text fields

Example:

Detect keywords such as:

- “biodegradable”
- “plastic-free”
- “recyclable”

Benefits:

- Automated ESG tagging
- Improved classification accuracy
- Reduced manual intervention

11.3 Scenario Simulation Engine:

Proposal:

Introduce parameterised modelling:

Examples:

- Tax rate increase (e.g., £223 → £300)
- Plastic reduction scenarios
- Volume growth projections

Implementation:

- Dynamic parameter tables
- Power BI What-If parameters

Benefits:

- CFO-ready forecasting
- Strategic planning tool

11.4 Incremental Data Pipeline:

Proposal:

Replace full refresh with:

- Incremental loading
- Partitioned processing

Benefits:

- Improved performance
- Reduced compute cost
- Scalability for production

11.5 Advanced ESG Metrics:

Add:

- Carbon footprint estimation
- Packaging recyclability scores
- Supplier ESG scoring

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12. Conclusion (Expanded)

This project successfully demonstrates how a modern data engineering approach can be used to:

1. Integrate ESG Data into Transactional Systems:

By combining:

- Internal retail sales data
- External sustainability metadata

The project overcomes the traditional separation between:

- Financial data systems
- Environmental data systems

2. Quantify Regulatory Exposure:

The system translates:

- Plastic usage → Tax liability

This enables:

- Financial visibility
- Compliance readiness
- Risk assessment

3. Identify Business Risk and Priorities

The analysis highlights:

- High-risk categories
- ESG data gaps
- Operational inefficiencies

This allows the organisation to:

- Prioritise interventions
- Allocate resources effectively

4. Demonstrate the Role of Data Engineering in ESG:

This project shows that:

Data engineering is not just about pipelines — it is a critical enabler of regulatory compliance and sustainability analytics.